The endpoint case using $J+J-3J$ as the main final interval

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Theorem 1. *Let $A \subset \mathbb{T}$ be a union of two non-empty disjoint open intervals with*
$$\mu(A) = \frac{1}{5}.$$
If
$$A + A - 3 \cdot A \neq \mathbb{T},$$
then, up to translating A and replacing A by $-A$, we have
$$A = \left(0, \frac{1}{10}\right) \cup \left(\frac{1}{2}, \frac{3}{5}\right).$$
In particular, $2 \cdot A$ is contained in an interval of length $\frac{1}{5}$.

Proof. As before, the condition $A + A - 3 \cdot A = \mathbb{T}$ is invariant under translating A and replacing A by $-A$. We may therefore assume
$$A = I \cup J, \qquad I = (0, \alpha), \qquad J = (x, x + \beta),$$
where
$$0 < \alpha < x < x + \beta < 1, \tag{1}$$
$$\alpha \geq \beta, \tag{2}$$
$$x - \alpha \leq 1 - (x + \beta). \tag{3}$$
We have
$$\alpha + \beta = \frac{1}{5},$$
and (3) gives
$$x \leq \frac{1 + \alpha - \beta}{2} = 3\alpha + 2\beta. \tag{4}$$
We will prove that $A + A - 3 \cdot A = \mathbb{T}$ unless
$$\alpha = \beta = \frac{1}{10}, \qquad x = \frac{1}{2}. \tag{5}$$

This time we keep $J + J - 3 \cdot J$ as the main final interval and leave $I + J - 3 \cdot J$ out of the main covering argument. Define

$$\begin{aligned} R_1 &= I + I - 3 \cdot I = (-3\alpha, 2\alpha), \\ R_2 &= I + J - 3 \cdot I = (x - 3\alpha, x + \alpha + \beta), \\ R_3 &= J + J - 3 \cdot I = (2x - 3\alpha, 2x + 2\beta), \\ R_4 &= I + I - 3 \cdot J = (-3x - 3\beta, -3x + 2\alpha), \\ R_6 &= J + J - 3 \cdot J = (-x - 3\beta, -x + 2\beta). \end{aligned}$$

All these intervals are in proper form. Clearly each of them is contained in $A + A - 3 \cdot A$.

The first part of the proof is exactly the same as before.

Claim 1. *If* (5) *does not hold, then*

$$x < 5\alpha \qquad \text{and} \qquad x < 4\alpha + \beta.$$

Proof. By (4) and $\alpha \geq \beta$,

$$x \leq 3\alpha + 2\beta \leq 5\alpha.$$

Equality forces $\alpha = \beta$ and $x = 3\alpha + 2\beta$, hence $\alpha = \beta = \frac{1}{10}$ and $x = \frac{1}{2}$. Thus, outside (5), we have $x < 5\alpha$.

Similarly,

$$x \leq 3\alpha + 2\beta \leq 4\alpha + \beta,$$

and equality again forces (5). Hence $x < 4\alpha + \beta$. □

Claim 2. *Assume* (5) *does not hold. Then*

$$R_1 \cup R_2 \cup R_3$$

covers $[0, 2x + 2\beta)$ *and also covers* $(1 - 3\alpha, 1)$.

Proof. Modulo 1, the interval $R_1 = (-3\alpha, 2\alpha)$ covers

$$[0, 2\alpha) \cup (1 - 3\alpha, 1).$$

By Claim 1, $x < 5\alpha$, so R_2 contains 2α. Therefore $R_1 \cup R_2$ covers $[0, x + \alpha + \beta)$. Again by Claim 1, $x < 4\alpha + \beta$, so R_3 contains $x + \alpha + \beta$. Therefore $R_1 \cup R_2 \cup R_3$ covers $[0, 2x + 2\beta)$. □

Thus the only possible leftover is

$$[2x + 2\beta, 1 - 3\alpha]. \tag{6}$$

If this interval is empty, we are done. So assume it is non-empty. Equivalently,

$$2x \leq 2\alpha + 3\beta. \tag{7}$$

We now cover this leftover using $R_4 \cup R_6$.

Claim 3. *The intervals R_4 and R_6 cover the leftover interval* (6)*, except possibly in the case*

$$\alpha = \frac{9}{65}, \qquad \beta = \frac{4}{65}, \qquad x = \frac{3}{13}. \tag{8}$$

In that case, the only point of (6) *not covered by $R_4 \cup R_6$ is $\frac{38}{65}$.*

Proof. Shift the leftover interval by -1. We need to cover

$$[2x + 2\beta - 1, -3\alpha].$$

The left endpoint is strictly to the right of the left endpoint of R_4, because

$$-3x - 3\beta < 2x + 2\beta - 1$$

is equivalent to $x + \beta > \frac{1}{5}$, which follows from $x > \alpha$ and $\alpha + \beta = \frac{1}{5}$.

The right endpoint is strictly to the left of the right endpoint of R_6, because

$$-3\alpha < -x + 2\beta$$

is equivalent to $x < 3\alpha + 2\beta$. This is strict under the present assumption that (6) is non-empty: indeed, equality in $x \leq 3\alpha + 2\beta$ together with (7) would give $6\alpha + 4\beta \leq 2\alpha + 3\beta$, impossible.

It remains only to check that R_4 and R_6 do not leave a gap inside this shifted leftover interval. Their adjacent endpoints are

$$-3x + 2\alpha \qquad \text{and} \qquad -x - 3\beta.$$

By (7),

$$-x - 3\beta \leq -3x + 2\alpha.$$

Thus R_4 and R_6 overlap, except possibly when

$$2x = 2\alpha + 3\beta.$$

If this inequality is strict, there is no gap and the whole leftover interval is covered.

Suppose then that $2x = 2\alpha + 3\beta$. In this case the leftover interval (6) is a single point, namely $1 - 3\alpha$. The only point not covered by $R_4 \cup R_6$ is the common endpoint

$$-3x + 2\alpha = -x - 3\beta.$$

This coincides with $(1 - 3\alpha) - 1 = -3\alpha$ precisely when

$$-3x + 2\alpha = -3\alpha,$$

or equivalently $3x = 5\alpha$. Combining this with $2x = 2\alpha + 3\beta$ gives

$$4\alpha = 9\beta.$$

Since $\alpha + \beta = \frac{1}{5}$, this gives

$$\alpha = \frac{9}{65}, \qquad \beta = \frac{4}{65}, \qquad x = \frac{3}{13}.$$

In that case the leftover point is

$$1 - 3\alpha = \frac{38}{65}.$$

This proves the claim. $\square$

The fake case (8) is not a genuine solution. It is exactly the case where the strategy using only $R_4 \cup R_6$ leaves an artificial endpoint. The omitted interval

$$R_5 = I + J - 3 \cdot J = (-2x - 3\beta, \alpha + \beta - 2x)$$

contains it, because in the fake case

$$R_5 = \left(-\frac{42}{65}, -\frac{17}{65}\right)$$

and

$$\frac{38}{65} - 1 = -\frac{27}{65} \in R_5.$$

Thus this point belongs to $A + A - 3 \cdot A$ after all.

We conclude that, outside the exceptional case (5), the set $A + A - 3 \cdot A$ is all of $\mathbb{T}$. Therefore, if $A + A - 3 \cdot A \neq \mathbb{T}$, we must be in the exceptional case (5).

In that case,

$$A = \left(0, \frac{1}{10}\right) \cup \left(\frac{1}{2}, \frac{3}{5}\right),$$

and

$$2 \cdot A = \left(0, \frac{1}{5}\right),$$

which is contained in an interval of length $\frac{1}{5}$. Directly computing the six interval pieces shows that this exceptional set satisfies

$$A + A - 3 \cdot A = \mathbb{T} \setminus \left\{\frac{1}{5}, \frac{7}{10}\right\}.$$

This completes the proof. $\square$

Remark 1. *The proof above shows that one cannot literally omit $I + J - 3 \cdot J$ from a covering proof in the open-interval setting. The five intervals R_1, R_2, R_3, R_4, R_6 leave a fake endpoint in the parameter choice*

$$\alpha = \frac{9}{65}, \qquad \beta = \frac{4}{65}, \qquad x = \frac{3}{13},$$

and that endpoint is covered exactly by the omitted term $I + J - 3 \cdot J$. Thus R_6 can be used as the main replacement for R_5, but R_5 still has to be mentioned to dismiss this one artificial endpoint.